(ii) $Var(X) = E(X^2) - \left(E(X)\right)^2$

Proof

We know $E(x) = \mu$

$$\begin{aligned} Var(X) &= E(X-\mu)^2 \\ &= E(X^2 - 2X\mu + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 \qquad \text{(Since } \mu \text{ is a constant)} \\ &= E(X^2) - 2\mu\mu + \mu^2 = E(X^2) - \mu^2 \\ Var(X) &= E(X^2) - \left(E(X)\right)^2 \end{aligned}$$

An alternative formula to compute variance of a random variable X is

$$\sigma^2 = \text{Var}(X) = E(X^2) - \left(E(X)\right)^2$$

(iii) $Var(aX+b) = a^2 Var(X)$ where a and b are constants

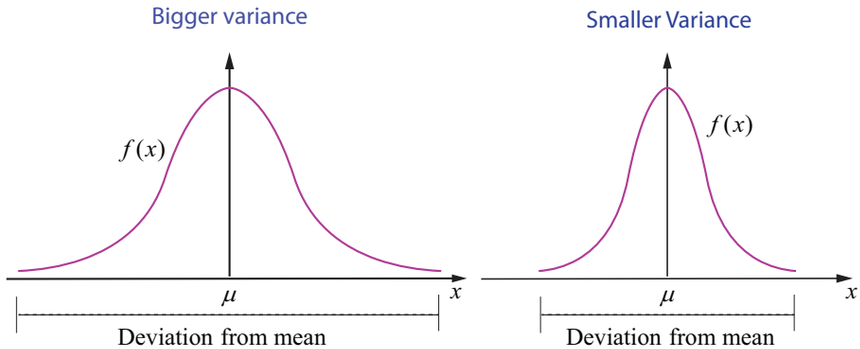
Proof

$$\begin{aligned} Var(aX+b) &= E\left((aX+b) - E(aX+b)\right)^2 \\ &= E\left(aX + b - aE(X) - b)\right)^2 \\ &= E\left(aX - aE(X)\right)^2 \\ &= E\left(a(X - E(X))\right)^2 \\ &= a^2 E\left(X - E(X)\right)^2. \end{aligned}$$

Hence $Var(aX+b) = a^2 Var(X)$

Corollary 3:	$V(aX) = a^2 V(X)$	(when $b = 0$)
Corollary 4:	$V(b) = 0$	(when $a = 0$)

Variance gives information about the deviation of the values of the random variable about the mean μ. A smaller σ^2 implies that the random values are more clustered about the mean, similarly, a bigger σ^2 implies that the random values are more scattered from the mean.



Different variance with same mean

Fig. 11.18

Chapter 11 Probability Distributions-new.indd 206 29-05-2020 13:07:29